JALIL HUSEYNOV

Mobile: +65 8916 8754 Email: jalil.huseynov@u.nus.edu https://github.com/jachelil https://jachelil.github.io/

EDUCATION

National University of Singapore **Aug 2023 - May 2027**
Bachelor of Computing (Honours) in Computer Science
- GPA: 4.79/5.00 (Highest Distinction)
- 2 x Dean's List - AY24/25 Sem 2, AY25/26 Sem1
- Second Major in Mathematics
- Relevant Courses: Optimisation Algorithms, Parallel & Distributed Algorithms, Algorithmic Mechanism Design, Combinatorics, Graph Theory, Theory of Computation, Mathematical Logic

RESEARCH EXPERIENCE

National University of Singapore *Undergraduate Researcher* **May 2026 - Present**
- Designing adaptive thresholds for a proof-of-useful-bandwidth blockchain secure against persistent adversaries, advised by Assoc. Prof. Haifeng Yu

National University of Singapore *Undergraduate Researcher* **Aug 2025 - May 2026**
- Formulated the Entropic Nearly Linear Freiman-Ruzsa Theorem (ENLFR), proving its equivalence to Entropic PFR and extending the 2023 Gowers-Green-Manners-Tao PFR resolution (advised by Assoc. Prof. Divesh Aggarwal)
- Proved a structural lower bound on the theorem via an explicit counterexample construction
- Applied entropy inequalities to translate combinatorial proof techniques into an information-theoretic setting

NUS Deep learning lab *Undergraduate Researcher* **Aug 2024 - Apr 2025**
- Collaborated in a team of 3 on optimising large language models (LLMs) for low-resource languages without requiring additional data
- Developed innovative methods to address multilingual model inconsistencies and enhance cross-lingual generalisation, contributing to a co-authored research paper on LLM optimisation (arXiv:2507.18918)

INTERNSHIP & WORK EXPERIENCE

Research in Industrial Projects for Students (RIPS) (Singapore), *Research Intern* **May 2026 - Jul 2026**
- Built an LLM-based (GPT-4o) pipeline for automated knowledge graph completion on smart-building data, sponsored by Kajima Corporation, and benchmarked it against 3 other ML approaches under class imbalance
- Fixed data pipeline bugs, migrated inference infrastructure, and co-authored the final report and sponsor presentation

Mercor *Math Proofs Expert - AI Model Evaluation* **Apr 2025 - Dec 2025**
- Evaluated LLM performance on advanced mathematical problem solving for a top-tier AI research organisation (NDA-protected)
- Reviewed, graded and corrected over 1,200 LLM-generated solutions to math problems, identifying subtle errors in mathematical logic and reasoning

National University of Singapore *Teaching Assistant* **Aug 2024 - Present**
- Served as Teaching Assistant (TA) for Linear Algebra, Introduction to AI & ML, Parallel & Distributed Algorithms, and Distributed Systems courses
- Created challenging quiz questions, graded weekly programming assignments of 15-20 students, organised weekly consultation sessions for 150+ students

ADDITIONAL POSITIONS HELD

Ministry of Education of Azerbaijan Republic, *Math Olympiad Leader, Coordinator and Coach* **Aug 2023 - May 2025**
- Coached national team for Junior Balkan Mathematics Olympiad 2024, achieving 2nd place among 20+ countries (1 Gold, 5 Silver medals)
- Led and mentored national team at International Mathematics Summer Camp (IMSC 2024), strengthening problem-solving and team preparation
- Authored original geometry problems adopted by top-tier competitions; selected for the Junior Balkan MO 2023 Shortlist and featured as the most challenging problem of the 2024 Azerbaijan National Olympiad

AWARDS

International and Regional Contests

- International Mathematical Olympiad (IMO) - 2x Bronze Medals (2023, 2022)
- Asian Pacific Mathematical Olympiad (APMO) - Silver Medal (2023)
- Junior Balkan Mathematical Olympiad (JBMO) - Gold Medal (2021)
- Azerbaijan National Mathematical Olympiad - Rank 1 nationally (2023, 2022, 2021)

SKILLS

- Technical Skills: Python, Java, C, PyTorch, Git, LaTeX, MATLAB
- Languages: English, Turkish, Azerbaijani